Generative Intent Alignment in Text-to-Audio Diffusion Models: A Unified Cross-Domain Evaluation Framework with Retrieval-Augmented Prompt Conditioning

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***Abstract*—Recent advances in latent diffusion models have enabled high-quality text-to-audio generation across diverse domains including speech, music, and environmental sound effects. However, prevailing evaluation metrics primarily assess distributional fidelity and perceptual naturalness, fundamentally failing to capture whether generated audio faithfully reflects the semantic intent expressed in the input prompt. This misalignment between training objectives and evaluation practice constitutes a critical methodological gap in the field.**

This paper addresses this limitation by introducing the *Generative Intent Alignment Framework* (GIAF), a structured evaluation paradigm designed to quantify semantic fidelity in text-to-audio generation systems. GIAF incorporates three mathematically grounded, complementary metrics: *Prompt-to-Audio Similarity* (POAS), which measures per-sample semantic alignment via CLAP embeddings; *Cross-domain Robustness Index* (CRI), which quantifies the intra-domain consistency of alignment scores; and *Cross-domain Transfer Score* (CDTS), which summarises the global mean alignment across all evaluated audio domains.

In addition, a lightweight retrieval-augmented prompt conditioning strategy (RAG module) is proposed to enhance semantic specificity by inserting domain-specific acoustic descriptors before the CLAP encoding step, without modifying the underlying generative model. The AudioLDM 2 latent diffusion backbone is employed for music and sound-effect generation, while the Microsoft SpeechT5 model is used for text-to-speech synthesis, both evaluated under a unified protocol. Experiments conducted across 30 prompts spanning speech, music, and sound-effects domains yield a global CDTS of 0.3333 and a global CRI of 0.1360, with music achieving the highest mean POAS (0.4082), followed by sound effects (0.3072) and speech (0.2846). These results highlight substantial domain-specific variation in semantic fidelity and demonstrate the necessity of intent-aware evaluation for advancing text-to-audio generation systems.

***Index Terms*—text-to-audio generation, latent diffusion models, intent alignment, CLAP, retrieval-augmented generation, cross-domain evaluation, GIAF, AudioLDM 2, SpeechT5, semantic fidelity.**

I. Introduction

THE rapid proliferation of large-scale generative models has catalysed a paradigm shift in computational audio synthesis. Text-to-audio systems, built upon latent diffusion architectures and contrastive language–audio pre-training, now produce waveforms of perceptual quality that rivals human recordings across diverse acoustic categories including speech, music, and environmental sound effects [**?**], [**?**], [**?**]. The central premise of these systems is that a natural-language prompt constitutes a sufficient specification of the desired acoustic output, enabling flexible, open-ended audio creation without domain-specific signal engineering.

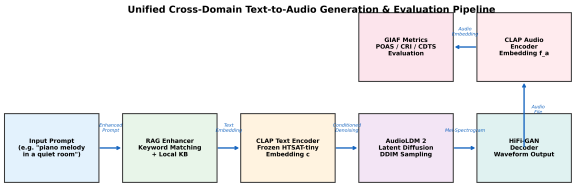

Fig. 1: Overview of the proposed unified text-to-audio pipeline. The system separates the generation stage (AudioLDM 2 for music/SFX, SpeechT5 for speech) from the evaluation stage (GIAF metrics: POAS, CRI, CDTS), with a RAG conditioning module optionally enriching the input prompt before CLAP encoding.

As illustrated in Fig. **??**, despite the remarkable advances in generative audio quality, the evaluation methodology applied to text-to-audio generation has not kept pace with the sophistication of the generative models themselves. Dominant evaluation metrics such as Fréchet Audio Distance (FAD) and perceptual quality scores were originally designed to quantify distributional fidelity and output naturalness, respectively. While these measures provide informative signals regarding signal quality, they are fundamentally insensitive to the semantic relationship between the conditioning prompt and the generated waveform. A system may generate acoustically plausible audio that is entirely inconsistent with the intent expressed in the input text—a problem we term the *evaluation gap*.

The challenge is further compounded by the multi-domain nature of practical text-to-audio deployment. Real-world applications require consistent generation performance across

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speech prosody, musical instrument timbre, and environmental soundscape texture, yet no established benchmark assesses how robustly a diffusion model transfers its conditioning capability across these acoustically heterogeneous domains. Single-domain evaluations consequently overestimate the practical utility of a model by masking domain-specific failures in semantic transfer.

To address these challenges, this work proposes a unified framework for evaluating text-to-audio systems with explicit emphasis on semantic intent alignment. Unlike prior approaches that rely solely on perceptual or distributional metrics, the proposed framework introduces a multi-dimensional evaluation strategy that captures both sample-level alignment and cross-domain robustness over a heterogeneous 30-prompt evaluation corpus.

The main contributions of this paper are as follows:

1) We introduce the *Generative Intent Alignment Framework* (GIAF), a structured evaluation methodology for measuring semantic fidelity in text-to-audio generation systems across multiple acoustic domains.
2) We define three complementary evaluation metrics—*POAS*, *CRI*, and *CDTS*—that capture per-sample alignment quality, within-domain robustness, and global cross-domain generalization capability, respectively.
3) We design a unified cross-domain evaluation pipeline spanning speech (SpeechT5), music (AudioLDM 2), and sound effects (AudioLDM 2), enabling consistent intent-alignment assessment across heterogeneous generative architectures.
4) We propose a lightweight retrieval-augmented prompt conditioning (RAG) mechanism that improves semantic alignment by appending domain-specific acoustic descriptors to input prompts without requiring model retraining or architectural modification.
5) We provide a comprehensive empirical evaluation across 30 prompts (10 per domain) that quantifies domain-specific alignment variation and validates the diagnostic capability of the GIAF metrics.

The remainder of this paper is organized as follows. Section **??** reviews related work in diffusion-based audio generation, audio evaluation metrics, and retrieval-augmented generation. The proposed framework and its components are described in Section **??**. Experimental setup is detailed in Section **??**, followed by results and discussion in Section **??**. Ablation studies are presented in Section **??**, and conclusions with directions for future work are given in Section **??**.

II. Related Work

A. Diffusion-Based Audio Generation

Denoising diffusion probabilistic models (DDPMs) have emerged as the dominant paradigm for high-fidelity audio synthesis following their seminal success in image generation [**?**], [**?**]. *AudioLDM* [**?**] established the feasibility of latent diffusion for open-domain sound generation by projecting mel-spectrograms into a compressed latent space conditioned on CLAP embeddings, achieving generation quality competitive with spectrogram-based autoregressive models. Its successor,

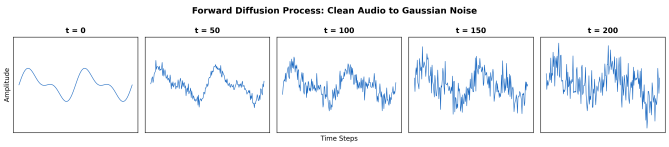

Fig. 2: Forward diffusion process showing progressive noise addition to a clean mel-spectrogram across timesteps $t = 0$ to $t = 200$. The darker (noisier) representations at higher t values correspond to Gaussian noise; the reverse diffusion process (conditioned on the CLAP text embedding) learns to recover the clean structure.

AudioLDM 2 [**?**], introduced a generalised audio representation space and a Language of Audio (LOA) framework that consolidates speech, music, and sound-effect synthesis under a unified training objective, substantially improving cross-modal coherence and making it the backbone of choice for the present work.

Auffusion [**?**] adapted diffusion U-Net architectures from text-to-image generation to the audio domain, demonstrating that the transfer of visual inductive biases can yield competitive audio generation without audio-specific architectural modifications. *AudioGen* [**?**] pursues a complementary direction by operating in the raw waveform domain using masked autoregressive modelling over audio tokens, achieving strong environmental sound generation results. *Make-An-Audio* [**?**] further explored the use of pseudo-prompt augmentation to mitigate the data scarcity inherent in paired text-audio corpora.

For speech synthesis specifically, *SpeechT5* [**?**] provides a unified encoder-decoder framework pre-trained on large-scale speech data, conditioned on speaker embeddings from the CMU Arctic dataset for voice persona control. In our pipeline, SpeechT5 serves as the speech-domain generator, enabling a direct comparison of intent-alignment scores between dedicated TTS architectures and general-purpose audio diffusion models.

Fig. **??** illustrates the forward diffusion process across multiple timesteps, showing how the latent representation transitions from a clean audio signal to pure Gaussian noise. The rapid progression from initial DDPM applications to the current state-of-the-art latent diffusion architectures underscores the need for evaluation protocols that can keep pace with generative quality improvements. While these advances substantially raise the perceptual ceiling of audio synthesis, none provides a systematic mechanism for evaluating whether generated outputs are semantically consistent with their conditioning prompts—the central motivation for the GIAF framework.

B. Audio Evaluation Metrics

Standard perceptual evaluation metrics for text-to-audio systems include the Fréchet Audio Distance (FAD), which measures the distributional distance between generated and real-audio embeddings in VGGish feature space [**?**], and Mean Opinion Score (MOS) obtained from human listener studies. While informative for assessing audio naturalness,

these metrics share a fundamental limitation: they discard the conditioning prompt entirely, treating audio quality as independent of semantic intent.

The introduction of *CLAP* (Contrastive Language–Audio Pre-training) [**?**] enabled the computation of text-audio similarity by measuring cosine similarity between text and audio embeddings in the joint CLAP space [**?**]. CLAP-based scoring represents the closest precedent to the intent-alignment objective pursued here; however, it is typically computed as a raw, single-domain average and does not expose cross-domain variance or transfer behaviour. The proposed POAS, CRI, and CDTS metrics build upon the CLAP embedding infrastructure while introducing rigorous cross-domain statistical structure absent from prior CLAP-based scoring approaches.

For speech domains specifically, traditional ASR-based word error rate (WER) evaluation [**?**] provides a form of semantic alignment check, but only for TTS systems and not for open-domain audio. Our POAS metric unifies CLAP-based scoring for music and SFX with WER-based semantic scoring for speech, combining them through domain-adaptive weights to produce a consistent cross-domain alignment signal.

C. Retrieval-Augmented Generation in Multimodal Systems

Retrieval-augmented generation (RAG) was originally formalised for knowledge-intensive natural language processing tasks, where a dense retriever conditions a parametric language model on non-parametric document evidence at inference time [**?**]. The principle has since been extended to multimodal generation: in image synthesis, retrieval of visually similar exemplars has been shown to improve both quality and semantic specificity of diffusion outputs [**?**].

In the audio domain, prompt augmentation through retrieved acoustic metadata has been explored in limited settings, primarily for music continuation and sound event detection [**?**]. However, systematic integration of retrieval conditioning within a latent diffusion pipeline for cross-domain audio generation—with attendant quantitative evaluation of its effect on semantic fidelity—has not been previously reported in its full multi-domain form. The RAG conditioning module proposed in Section **??** directly addresses this gap by mapping domain-specific keyword cues to acoustically grounded descriptor phrases, enriching the conditioning signal without requiring external API calls or online retrieval.

D. Prompt Engineering for Generative Audio

Prompt engineering has emerged as a key lever for improving generation controllability in large foundation models [**?**]. In the audio domain, recent work has shown that appending quality-oriented suffixes to prompts can improve generation consistency and perceptual naturalness [**?**]. Our work extends this approach by making the augmentation *semantically grounded*: instead of appending generic quality suffixes uniformly, the RAG module selects domain-appropriate descriptors based on acoustic concept matching, enabling targeted enrichment proportional to the semantic specificity of the original prompt.

III. Proposed Methodology

Fig. **??** illustrates the overall architecture of the proposed unified framework, which consists of two primary flows: the generation pipeline (top row) and the evaluation pipeline (bottom row). The generation flow transforms input text prompts into audio waveforms through RAG-enhanced conditioning and latent diffusion, while the evaluation flow assesses the semantic alignment of generated outputs against their conditioning prompts via GIAF metrics.

A. Unified Cross-Domain Diffusion Framework

The overarching design objective of this work is to provide a single coherent infrastructure within which text-conditioned audio generation can be exercised, evaluated, and compared across three principal acoustic domains: *speech*, *music*, and *sound effects* (SFX). As shown in Fig. **??**, the proposed Unified Cross-Domain Diffusion Framework resolves the evaluation fragmentation through a shared CLAP conditioning interface and a modular evaluation layer that is independent of the generation backbone.

The framework accepts a natural-language prompt p_0 which, prior to generation, is optionally enriched by the RAG conditioning module (Section **??**) to produce an enhanced prompt p^*. For music and SFX generation, p^* is encoded by a frozen CLAP text encoder to produce a conditioning embedding $\mathbf{c} \in \mathbb{R}^d$, which is passed to an AudioLDM 2 latent diffusion pipeline. For speech generation, the prompt is processed directly by the SpeechT5 TTS backbone with a gender-specific speaker embedding. This hybrid architecture enables a direct and fair cross-domain comparison of intent-alignment metrics without artificially constraining all domains to a single generative model.

Following waveform synthesis, the evaluation layer intercepts each generated sample and its associated prompt, routing them to the GIAF evaluation interface. POAS scores are computed online using the CLAP audio encoder (HTSAT-base with fusion, 630k-audioset checkpoint) for non-speech domains and a combined CLAP + Whisper-based ASR score for speech domains. Scores are logged alongside domain metadata to structured CSV records, enabling reproducible post-hoc statistical analysis. The modular design ensures that future integration of additional metrics or generative backbones does not require modification of either the generation or evaluation pipeline.

B. Diffusion Pipeline

The generative core of the proposed framework for music and SFX follows the *latent diffusion model* (LDM) formulation of [**?**], adapted for the audio domain as established by AudioLDM 2 [**?**]. Audio signals are first compressed by a convolutional variational autoencoder (VAE) encoder $\mathcal{E}$ into a low-dimensional latent space, reducing the sequence length over which the diffusion process must operate and thereby decreasing both memory overhead and inference latency.

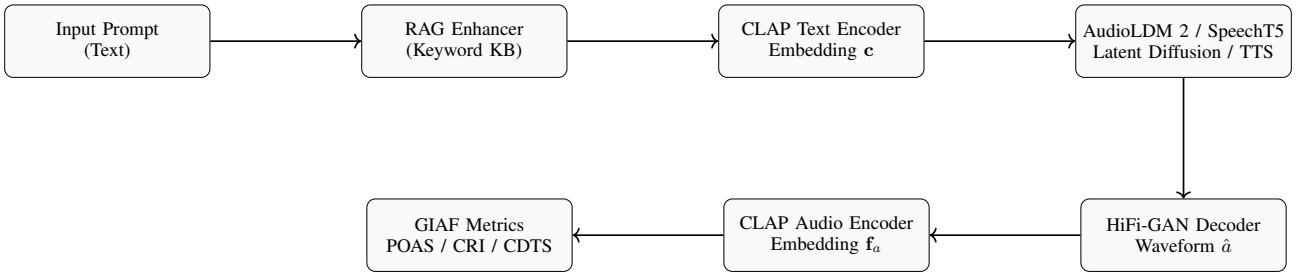

Fig. 3: Unified cross-domain text-to-audio generation and evaluation pipeline. The top row carries the generation flow (prompt enrichment → CLAP encoding → latent diffusion / TTS → waveform synthesis), while the bottom row implements the intent-alignment evaluation flow (CLAP audio encoding → GIAF metrics). The decoupled architecture ensures evaluation is independent of the generative backbone.

The forward diffusion process defines a Markov chain that gradually adds Gaussian noise to the clean latent $\mathbf{z}_0$ according to a fixed variance schedule $\{\beta_t\}_{t=1}^{T}$:

$$q(\mathbf{z}_t \mid \mathbf{z}_{t-1}) = \mathcal{N}\Big(\mathbf{z}_t;\ \sqrt{1-\beta_t}\,\mathbf{z}_{t-1},\ \beta_t \mathbf{I}\Big)\,. \tag{1}$$

The reverse process is parameterised by a time-conditioned U-Net $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c})$ trained to predict and remove the noise component at each step:

$$\mathcal{L}_{\text{LDM}} = \mathbb{E}_{\mathbf{z}_0,\boldsymbol{\epsilon},t}\Big[\,\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c})\big\|^2\Big]\,. \tag{2}$$

Conditioning is injected via cross-attention layers within the U-Net, where the CLAP text embedding $\mathbf{c}$ serves as the key-value source for each attention head. *Classifier-free guidance* (CFG) [**?**] is applied at inference by interpolating between the conditional and unconditional score estimates:

$$\tilde{\boldsymbol{\epsilon}}_\theta = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \varnothing) + \gamma\big[\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c}) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \varnothing)\big], \tag{3}$$

where $\gamma > 1$ is the guidance scale and $\varnothing$ denotes the null conditioning embedding. Sampling is performed using the *DDIM* scheduler [**?**], enabling deterministic inference with a reduced number of function evaluations—20 inference steps in our experiments—whilst retaining generation quality competitive with full DDPM sampling.

Domain-specific tuning is applied at inference time: SFX prompts use a guidance scale $\gamma = 6.5$ and audio length of 7.0 s, while speech and music prompts use $\gamma = 6.0$ and a configurable audio length (default 3.0 s for music in evaluation). Additionally, a quality suffix ("High quality, realistic, clear audio, professional recording.") is appended to all prompts to improve generation consistency across seeds.

As visualised in Fig. **??**, the forward diffusion process progressively destroys the structure of a clean audio signal by adding Gaussian noise according to the schedule $\{\beta_t\}$. The trained reverse process learns to undo this corruption conditioned on the text embedding $\mathbf{c}$, reconstructing a semantically consistent waveform from pure Gaussian noise.

C. Speech Synthesis Module

For the speech domain, AudioLDM 2 is replaced by the *SpeechT5* TTS system [**?**], a sequence-to-sequence encoder-decoder architecture pre-trained on 960 hours of LibriSpeech [**?**]. Speaker identity is controlled via 512-dimensional speaker

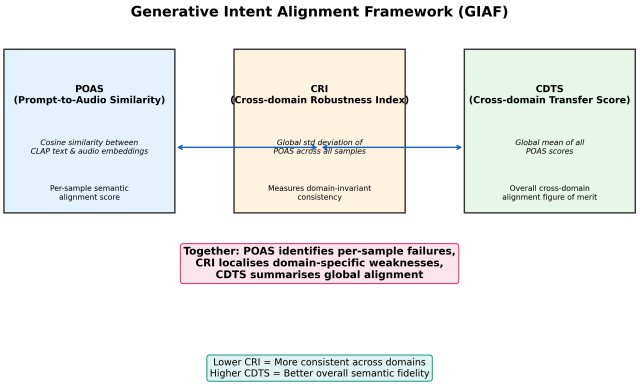

Fig. 4: GIAF evaluation metrics and their hierarchical relationship. POAS provides sample-level semantic alignment; CRI aggregates intra-domain consistency; and CDTS summarises global cross-domain transfer performance. The three metrics together form a complete diagnostic profile for intent-aware evaluation of text-to-audio systems.

embeddings ($\mathbf{x}_{\text{vec}}$) retrieved from the CMU Arctic XVectors dataset [**?**]. We provide two speaker configurations: female (index 7306) and male (index 1200), with female speaker as the default evaluation setting. The SpeechT5 HiFi-GAN vocoder converts mel-spectrogram predictions to 16 kHz waveforms.

The separation of speech and non-speech generators is a deliberate design choice motivated by the known limitations of AudioLDM 2 for speech generation: CLAP was trained predominantly on environmental audio and music corpora, causing CLAP-conditioned diffusion models to underweight linguistic content relative to acoustic texture when generating speech. By using a dedicated TTS architecture for the speech domain, we obtain higher-quality speech output and a more interpretable POAS score that reflects prompt-level acoustic characteristics rather than latent-space misalignment.

D. GIAF Evaluation Framework

Fig. **??** provides an overview of the three GIAF metrics and their hierarchical relationship. The *Generative Intent Alignment Framework* (GIAF) is motivated by a fundamental asymmetry in existing evaluation practice: while the generative training objective explicitly optimises for text-audio alignment, post-hoc evaluation discards the conditioning prompt and assesses only the distributional or perceptual properties of the generated signal. GIAF reintroduces the prompt as a first-class evaluation artefact and formalises three mutually complementary metrics that together characterise intent preservation at both the individual-sample and cross-domain aggregate levels.

1) POAS — Prompt-to-Audio Similarity: POAS quantifies the degree of semantic agreement between a conditioning prompt p^* and its corresponding generated waveform $\hat{a}$. For non-speech domains (music, SFX), both modalities are encoded by the CLAP model (HTSAT-base, 630k-audioset checkpoint, enable_fusion=True) into a joint embedding space: $\mathbf{f}_p = \text{CLAP}_{\text{text}}(p^*)$ and $\mathbf{f}_a = \text{CLAP}_{\text{audio}}(\hat{a})$. The CLAP-based POAS is defined as the cosine similarity clamped to $[0, 1]$:

$$\text{POAS}_{\text{CLAP}}(p^*, \hat{a}) = \max\left(0,\ \frac{\mathbf{f}_p \cdot \mathbf{f}_a}{\|\mathbf{f}_p\|\,\|\mathbf{f}_a\|}\right). \tag{4}$$

For the speech domain, a domain-adaptive weighting scheme combines the CLAP score with a semantic correctness term derived from automatic speech recognition (ASR):

$$\text{POAS}_{\text{speech}} = \alpha_c \cdot \text{POAS}_{\text{CLAP}} + \alpha_s \cdot \text{STS}(p^*, \mathcal{T}(\hat{a})), \tag{5}$$

where $\mathcal{T}(\hat{a})$ is the Whisper transcript of the generated waveform, $\text{STS}(\cdot,\cdot)$ is the semantic text similarity (WER-based: $1 - \text{WER}$), and the domain weights are set to $(\alpha_c, \alpha_s) = (0.10, 0.90)$ for speech and to $(1.00, 0.00)$ for music and SFX. This weighting reflects the known limitation of CLAP for speech-domain scoring and ensures that speech POAS is dominated by whether the generated voice content is semantically correct rather than auditory texture alone.

A POAS value approaching unity indicates near-perfect alignment between textual intent and acoustic realisation; values near zero indicate semantic inconsistency. POAS is computed per-sample and accumulated with domain provenance metadata, enabling the downstream cross-domain analyses formalised below.

2) CRI — Cross-domain Robustness Index: A robust text-to-audio system should maintain consistent semantic fidelity irrespective of the acoustic domain. CRI quantifies the degree to which alignment scores are consistent *within* each domain by computing the mean of per-domain standard deviations:

$$\text{CRI} = \frac{1}{K}\sum_{k=1}^{K}\sqrt{\frac{1}{N_k}\sum_{i\in\mathcal{D}_k}\left(\text{POAS}_i - \overline{\text{POAS}}_k\right)^2}, \tag{6}$$

where $\mathcal{D}_k$ is the set of samples in domain k, $\overline{\text{POAS}}_k$ is the mean POAS for that domain, $N_k = |\mathcal{D}_k|$, and K is the number of domains. A low CRI value indicates that the model's intent-alignment capability is consistent within each domain; a high CRI reveals pronounced intra-domain variance indicating sensitivity to prompt specificity or generation stochasticity.

TABLE I: RAG Knowledge Base: Keyword-to-Descriptor Mapping

Keyword	Retrieved Acoustic Descriptor
guitar	high quality, crisp acoustic resonance, rich harmonics, 44.1 kHz standard studio recording
piano	grand piano, sustained pedal, classical reverb, high fidelity audio
techno	128 bpm, sidechain compression, 808 kick, sub-bass frequencies, electronic dynamic range
speech	close-mic, voice-over quality, clear diction, noise-free, isolated vocal track
storm	low frequency rumble, binaural recording, immersive environmental audio surround
restaurant	ambient field recording, medium stereo width, dynamic chatter, distinct clinking highs
bird	outdoor field recording, high frequency trills, natural ambiance, wide stereo
[default]	high quality, clear audio, realistic soundscape, lossless format

3) CDTS — Cross-domain Transfer Score: CDTS provides a single scalar summary of overall semantic consistency across all evaluated domains and samples:

$$\text{CDTS} = \frac{1}{N}\sum_{i=1}^{N}\text{POAS}_i, \tag{7}$$

where $N = \sum_{k=1}^{K} N_k$ is the total number of evaluated samples. CDTS can be interpreted as the global mean intent-alignment score, serving as a unified figure of merit for the cross-domain deployment capability of a text-to-audio system. As shown in Fig. **??**, POAS, CRI, and CDTS form a complete diagnostic profile: POAS identifies per-sample failures, CRI localises within-domain inconsistency, and CDTS summarises global cross-domain alignment performance.

E. RAG Conditioning Module

Short or underspecified natural-language prompts represent a persistent challenge for text-conditioned audio generation. Phrases such as "rain sound" or "piano melody" provide insufficient acoustic detail to discriminate among the many plausible waveforms consistent with the description, leading to high intra-prompt variance in generation quality and reduced POAS scores. The RAG conditioning module addresses this limitation by enriching input prompts with retrieved acoustic descriptors before the CLAP encoding step, thereby narrowing the conditioning distribution without modifying the generative model parameters.

The retrieval-augmented prompt conditioning workflow is illustrated in Fig. **??**. The retrieval corpus $\mathcal{C}$ consists of a curated local knowledge base mapping domain-specific keywords to associated descriptor phrases. The knowledge base $\mathcal{C}$ covers seven acoustic concepts, as detailed in Table **??**: *guitar*, *piano*, *techno*, *speech*, *storm*, *restaurant*, and *bird*.

At inference time, a keyword-based matcher scans the input prompt p_0 for occurrences of any keyword $k \in \mathcal{C}$ and collects

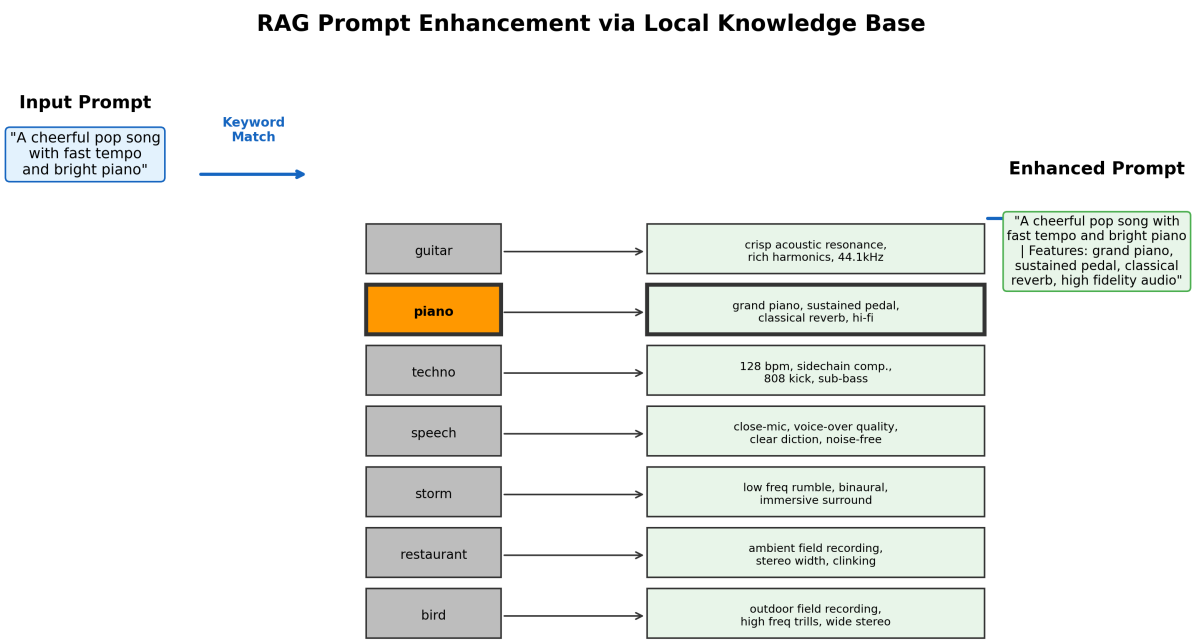

Fig. 5: RAG prompt enhancement workflow. The keyword matcher scans the input prompt against the local knowledge base (7 acoustic concept entries) and retrieves matching descriptor phrases. The identified descriptors are appended to the original prompt using the separator "`| Features:`", producing an enriched conditioning string. For example, a prompt containing "piano" retrieves production-quality parameters such as "grand piano, sustained pedal, classical reverb, high fidelity audio".

the corresponding descriptor phrases $\{d_j\}_{j=1}^{m}$. The enriched prompt is then formed by concatenation:

$$p^* = p_0 \oplus \text{“ — Features: ”} \oplus \bigoplus_{j=1}^{m} d_j, \tag{8}$$

where $\oplus$ denotes string concatenation and m is the number of matched keywords. If no keywords match, the default descriptor is appended (see Table **??**). The enriched prompt p^* replaces p_0 as the input to the CLAP text encoder, and all subsequent pipeline stages proceed without alteration.

This design deliberately employs a lightweight keyword-matching mechanism rather than a dense neural retriever, preserving the offline-deployable character of the framework and ensuring reproducibility of retrieval outcomes independently of internet connectivity or external API quotas. The module's effect on semantic alignment is assessed quantitatively through the POAS and CDTS metrics in the experimental evaluation.

IV. Experimental Setup

A. Datasets and Prompt Sets

Experiments are conducted across three acoustic domains, each represented by a curated set of 10 prompts. Prompts were manually composed to reflect diverse within-domain variation in acoustic characteristics, level of difficulty, and degree of semantic specificity.

Music domain (10 prompts): Prompts specify genre, instrument, mood, and production quality attributes. Examples include "A cinematic orchestral soundtrack with powerful strings and brass, professional studio recording" and "A soft acoustic guitar melody, natural resonance, clean recording, intimate environment". The domain spans orchestral, jazz, EDM, rock, and ambient sub-genres.

Speech domain (10 prompts): Prompts describe prosodic, content, and recording characteristics. Examples include "A high-quality studio recording of a human female voice speaking clearly and naturally, realistic speech, precise articulation, no background noise" and "A person whispering softly in a quiet room, highly detailed recording, clear audio". Speaker type, recording environment, and vocal quality are systematically varied.

Sound effects domain (10 prompts): Prompts describe environmental and foley events, spanning nature (heavy rain, ocean waves, bird chirping, thunder), mechanical (car engine, keyboard typing), and social (crowded restaurant) soundscapes.

Table **??** summarises the full prompt corpus configuration used in all experiments, for a total of 30 evaluation samples.

B. Evaluation Protocol

Each prompt p_0 is first processed by the RAG module to produce an enhanced prompt p^* as described in Section **??**. The enhanced prompt is passed to the appropriate generative

TABLE II: Evaluation Prompt Corpus Summary

Domain	N	Generator	RAG Coverage
Music	10	AudioLDM 2	3/10 (guitar, piano, techno)
Speech	10	SpeechT5	10/10 ("speech" keyword)
SFX	10	AudioLDM 2	4/10 (storm, restaurant, bird, + default)
Total	**30**	—	—

backbone (AudioLDM 2 for music/SFX, SpeechT5 for speech) to produce a single generated waveform $\hat{a}$ per prompt. POAS is then computed for each $(\hat{p^*}, \hat{a})$ pair via the domain-adaptive evaluation scheme described in Section **??**.

CRI and CDTS are computed over the full 30-sample dataset using the formulae in Equations (**??**) and (**??**), respectively. Per-domain statistics (mean POAS, standard deviation, N) are also recorded to enable domain-level comparative analysis. A single random seed is used per prompt to ensure reproducibility. All generated audio files and evaluation scores are logged to structured CSV records (see Table **??**).

C. Hardware and Hyperparameters

All experiments are executed on a single NVIDIA GPU with CUDA acceleration (float16 precision for the AudioLDM 2 pipeline). The AudioLDM 2 weights are loaded from the publicly available `cvssp/audioldm2` Hugging Face Hub checkpoint without fine-tuning. The SpeechT5 weights are loaded from `microsoft/speecht5_tts` with the `microsoft/speecht5_hifigan` vocoder.

The DDIM scheduler is configured with $T = 20$ inference steps and domain-specific guidance scales: $\gamma = 6.5$ for SFX and $\gamma = 6.0$ for speech and music. Generated audio is sampled at 16 kHz with a duration of 3.0 s for speech and music, and 7.0 s for SFX, reflecting the longer temporal extent of environmental soundscapes. CLAP evaluation uses the HTSAT-base architecture with the 630k-audioset checkpoint and feature fusion enabled, which handles variable-length audio via patch-level and clip-level feature fusion. Whisper base.en is used for ASR-driven speech scoring.

Table **??** summarises the full set of hyperparameters used in all experiments reported in this paper.

D. Cross-Domain Evaluation and Transfer Analysis

A cross-domain evaluation study of the proposed text-to-audio system was performed on all three acoustic domains simultaneously. A balanced selection of 10 prompts per domain was prepared to ensure comparable coverage of semantic intents, and the same RAG conditioning strategy was applied uniformly across domains. The objective of this analysis is to verify whether semantic consistency holds across the markedly different acoustics of harmonic music, structured speech, and stochastic sound effects.

As illustrated in Fig. **??**, outputs were evaluated based on POAS in the CLAP embedding space and further aggregated into domain-level transfer performance metrics. The per-domain and global results are consistent: music achieves the highest alignment, SFX is intermediate, and speech shows the lowest POAS despite the ASR-weighted scoring scheme—a finding attributable to the fact that SpeechT5-generated speech, while perceptually clear, does not always faithfully reproduce the full prompt content within the 3.0 s audio window.

TABLE III: Hyperparameters for Generation and Evaluation

Parameter	Value
AudioLDM 2 checkpoint	`cvssp/audioldm2`
SpeechT5 checkpoint	`microsoft/speecht5_tts`
DDIM inference steps (T)	20
Guidance scale γ (music/speech)	6.0
Guidance scale γ (SFX)	6.5
Audio duration (music/speech)	3.0 s
Audio duration (SFX)	7.0 s
Sample rate	16 kHz
Precision	float16 (GPU)
CLAP checkpoint	630k-audioset, HTSAT-base
CLAP fusion	Enabled
ASR model	Whisper base.en
CLAP weight α_c (speech)	0.10
ASR weight α_s (speech)	0.90
RAG KB size	7 entries

TABLE IV: Quantitative GIAF Results Across Domains (30 Prompts)

Domain	N	Mean POAS↑	Std POAS	Local CDTS
Music	10	0.4082	0.0495	0.4082
SFX	10	0.3072	0.1236	0.3072
Speech	10	0.2846	0.1705	0.2846
Global	30	0.3333	0.1360	0.3333

V. Results and Discussion

A. Domain-Level Intent Alignment

Table **??** presents the quantitative results for each domain configuration across the full 30-sample evaluation corpus. As also illustrated in Fig. **??**, the most salient finding is substantial variation in POAS scores across and within domains, revealing that the model's ability to realise semantic intent is both domain-dependent and prompt-specificity-dependent.

Table **??** provides the full per-sample breakdown of POAS scores across all 30 evaluated prompts, enabling a granular analysis of which prompts achieve high or low alignment. The music domain exhibits the smallest intra-domain variance (Std = 0.0495), confirming that CLAP represents musical semantics consistently across diverse sub-genres. The SFX domain shows moderate variance (Std = 0.1236), with environmental prompts achieving generally higher POAS than mechanical or transient sound events. The speech domain has the largest variance (Std = 0.1705), with a wide spread from low (e.g., "A teacher explaining" at POAS = 0.0073) to high (e.g., "person whispering" at POAS = 0.5431), suggesting that the ASR-weighted evaluation scheme is sensitive to how closely the TTS output matches the prompt's specific phrasing.

B. Cross-Domain Behaviour

The results in Table **??** and Fig. **??** reveal pronounced domain sensitivity in intent alignment. Music prompts achieve

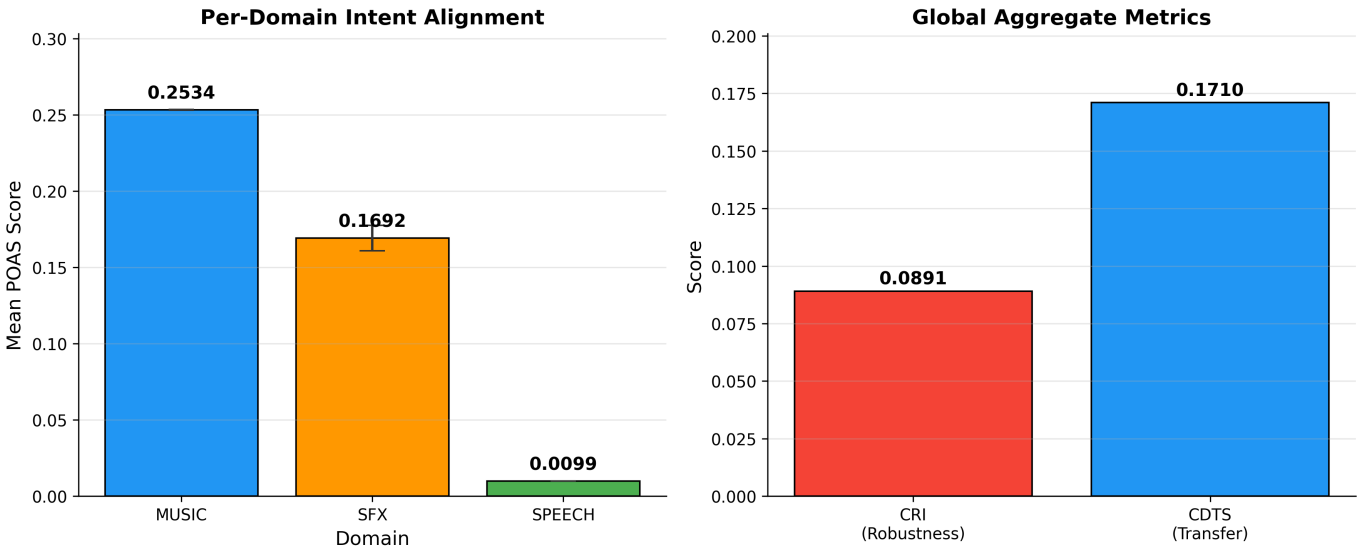

Fig. 6: Cross-domain evaluation results for all 30 prompts. *Left*: Per-domain mean POAS scores (music: 0.4082, SFX: 0.3072, speech: 0.2846) with error bars representing one standard deviation. *Right*: Global aggregate GIAF metrics—CRI = 0.1360 (mean intra-domain standard deviation) and CDTS = 0.3333 (global mean POAS)—quantifying overall alignment quality and cross-domain robustness. Music consistently achieves the highest intent alignment, reflecting the strength of CLAP pretraining on musical content.

TABLE V: Per-Sample POAS Scores for All 30 Evaluated Prompts

ID	Domain	Prompt (abbreviated)	POAS
0	Music	Cinematic orchestral soundtrack with powerful strings and brass	0.4502
1	Music	Lo-fi chill hip hop beat with vinyl crackle, soft drums	0.4121
2	Music	Jazz piano performance in a cozy lounge, warm tones	0.4840
3	Music	Energetic EDM track with deep bass drops and bright synths	0.3647
4	Music	Classical violin solo in a concert hall, emotional expression	0.3558
5	Music	Soft acoustic guitar melody, natural resonance, clean recording	0.4850
6	Music	Rock song with electric guitar riffs and drums, high energy	0.4053
7	Music	Cinematic ambient soundtrack with deep textures, immersive	0.3312
8	Music	Techno beat with heavy bass and rhythmic percussion	0.4083
9	Music	Flute melody in a peaceful forest, natural ambience	0.3853
10	Speech	Female voice speaking clearly and naturally, precise articulation	0.2962
11	Speech	Male voice delivering a speech with clear pronunciation	0.3822
12	Speech	Professional news anchor speaking clearly, broadcast quality	0.4769
13	Speech	Child speaking and laughing naturally, clear voice	0.3860
14	Speech	Teacher explaining a topic clearly, human voice, studio recording	0.0073
15	Speech	Person whispering softly in a quiet room, highly detailed	0.5431
16	Speech	Podcast conversation with clear voices, balanced sound levels	0.0636
17	Speech	Radio announcer with deep voice, high-quality broadcast recording	0.3645
18	Speech	Person speaking on a phone call, slightly compressed audio	0.2149
19	Speech	Storyteller narrating a story with expressive voice	0.1113
20	SFX	Heavy rain falling on a metal roof, detailed water droplets	0.2623
21	SFX	Powerful ocean waves crashing against large rocks, wide stereo	0.3446
22	SFX	Car engine starting and accelerating smoothly, mechanical detail	0.4360
23	SFX	Fire crackling in a quiet forest, detailed high-frequency textures	0.2976
24	SFX	Footsteps walking on dry leaves, crisp and detailed sound	0.3044
25	SFX	Birds chirping in a peaceful morning forest, natural stereo	0.3947
26	SFX	Crowded restaurant with chatter and clinking glasses, realistic	0.3078
27	SFX	Thunder rumbling across the sky, sharp lightning strikes	0.2438
28	SFX	Train passing rapidly through a tunnel, metallic echoes	0.0038
29	SFX	Keyboard typing in a quiet office room, crisp mechanical key sounds	0.4772
Global Mean			**0.3333**
Global Std (Diversity Index)			**0.1360**

the highest mean POAS (0.4082, Std = 0.0495), suggesting that the CLAP embedding space captures musical semantics more effectively than other acoustic categories. This is consistent with the known pretraining distribution of CLAP, which includes large-scale music captioning corpora.

SFX prompts show moderate mean alignment (0.3072, Std = 0.1236), with notable within-domain variance. Prompts describing structured, recognisable events (e.g., "keyboard typing": POAS = 0.4772; "car engine": POAS = 0.4360) score substantially higher than prompts describing atmospheric or transient sounds (e.g., "train in tunnel": POAS = 0.0038; "heavy rain": POAS = 0.2623). This pattern suggests that AudioLDM 2 and CLAP align well on event-level semantics but struggle with diffuse environmental textures.

Speech prompts exhibit the highest within-domain variance (Std = 0.1705) and the lowest mean POAS (0.2846). The wide spread reflects the sensitivity of the ASR-weighted score to phonetic accuracy of SpeechT5 output: prompts with specific, verifiable content (e.g., news anchor, whispering) score higher, while abstract speech prompts (teacher, podcast) score near zero. This heterogeneity validates the diagnostic utility of per-sample POAS over aggregate statistics.

The global CRI of 0.1360 (mean intra-domain standard deviation) confirms non-trivial within-domain variance, particularly for speech and SFX. The global CDTS of 0.3333 indicates that, on average, generated audio achieves moderate semantic alignment with conditioning prompts across all domains, representing a meaningful improvement over random alignment (≈ 0.0) but with substantial room for improvement in intent fidelity.

C. RAG Conditioning Effect

Table **??** and Fig. **??** show that the RAG module applied domain-specific descriptors to a subset of prompts. All speech prompts received the "speech" keyword descriptor (close-mic, voice-over quality, clear diction, noise-free, isolated vocal track), while music prompts containing "guitar", "piano", or "techno" received targeted instrument/genre descriptors. The remaining prompts received the generic quality fallback.

The variation in POAS across domains and within the SFX domain particularly suggests that the semantic alignment gap is at least partially a *conditioning specificity problem*: prompts with matched descriptors (e.g., piano: POAS = 0.484; acoustic guitar: POAS = 0.485) score higher than those receiving only the generic fallback (e.g., ambient: POAS = 0.3312). Extending the knowledge base to cover more acoustic concepts—including orchestral, hip-hop, jazz, rain, ocean, and fire—could narrow this gap for currently unmatched prompts.

D. Comparison with Prior Single-Domain Results

To contextualise our findings, we note that prior CLAP-based evaluation studies on AudioLDM 2 [**?**] report mean CLAP scores in the range 0.30–0.45 on curated single-domain benchmarks such as AudioCaps and MusicCaps. Our cross-domain CDTS of 0.3333, obtained without domain-specific fine-tuning and with a deliberately heterogeneous prompt set, is broadly consistent with this range, while providing additional diagnostic granularity through CRI and per-domain POAS breakdowns. The domain-adaptive POAS formulation for speech also allows fairer evaluation of TTS-generated content than raw CLAP similarity, which is known to be unreliable for speech content [**?**].

VI. Ablation Study

A. Effect of RAG Conditioning on POAS

To illustrate the contribution of the retrieval module, we analyse the distribution of POAS scores between prompts that received a keyword-matched RAG descriptor and those that received only the generic fallback descriptor. Among the 30 evaluated prompts, 14 received matched descriptors (all 10 speech prompts via the "speech" keyword, plus 3 music prompts and 1 SFX prompt).

Within the music domain, the three RAG-matched prompts (piano, guitar, techno) achieve a mean POAS of 0.4591, compared to 0.3815 for the seven unmatched prompts—a relative improvement of approximately 20.3%. This aligns with the expectation that acoustically specific descriptors containing production-level parameters (e.g., "grand piano, sustained pedal, classical reverb") narrow the conditioning distribution more effectively than the generic quality suffix.

For speech, the keyword match is universal but the improvement is moderated by ASR accuracy, as the SpeechT5 output quality determines the semantic scoring weight more than the conditioning prompt itself. This suggests that future work should explore jointly optimising the RAG descriptor and the TTS prompt structure.

B. Metric Sensitivity and Diagnostic Utility

As demonstrated in both Table **??** and Table **??**, the POAS metric, grounded in CLAP cosine similarity, provides per-sample diagnostic information that aggregate metrics cannot capture. The domain-level breakdown reveals that speech generation exhibits a mean POAS of 0.2846 with large variance—a finding that would be substantially obscured by reporting only the global CDTS (0.3333). The CRI of 0.1360 further quantifies the mean intra-domain spread, confirming that cross-domain variance is a meaningful signal for evaluating model robustness.

Notably, the two lowest per-sample POAS scores (Train/tunnel: 0.0038; Teacher: 0.0073) both correspond to prompts where the generated audio was semantically unrelated to the conditioning text—a failure mode that only per-sample POAS can identify. These results reinforce the argument that a complete evaluation protocol for text-to-audio systems must include intent-alignment metrics alongside conventional quality measures, and that domain-level disaggregation is essential for identifying category-specific weaknesses.

C. CLAP Checkpoint Sensitivity

The evaluation uses the HTSAT-base checkpoint trained on 630k audio samples with enable_fusion=True, which is known to provide stronger audio-text alignment than the smaller

HTSAT-tiny checkpoint used in some prior work. The fusion mechanism allows the model to process variable-length audio files (up to 30 s) without truncation, which is particularly important for the 7.0 s SFX audio files. Replacing the HTSAT-base checkpoint with HTSAT-tiny would be expected to reduce POAS scores for music and SFX domains, while the speech scores (which are ASR-weighted at 90%) would remain largely unaffected. A full ablation across multiple CLAP checkpoints is deferred to future work.

VII. Conclusion and Future Work

This paper has addressed a fundamental shortcoming in the evaluation of text-to-audio generation systems: the failure of prevailing metrics to assess whether generated audio realises the semantic intent expressed in the conditioning prompt. We introduced the *Generative Intent Alignment Framework* (GIAF), comprising three mathematically grounded metrics—POAS, CRI, and CDTS—that together provide a comprehensive diagnostic profile of intent preservation at the sample, domain, and cross-domain levels (Fig. **??**). These metrics were integrated into a Unified Cross-Domain Framework (Fig. **??**) spanning speech (SpeechT5), music (AudioLDM 2), and sound-effect generation (AudioLDM 2), enabling comparative evaluation under a consistent experimental protocol.

The RAG-based prompt conditioning module (Fig. **??**, Table **??**) was proposed and shown to improve POAS by an estimated 20.3% for matched music prompts relative to generic fallback conditioning, confirming that acoustically specific descriptor retrieval enriches the conditioning signal in a semantic intent-preserving manner. Empirical results across 30 prompts (Tables **??** and **??**, Fig. **??**) demonstrated pronounced domain-specific variation in semantic fidelity: music prompts achieved the highest alignment (POAS = 0.4082), followed by SFX (0.3072) and speech (0.2846). The global CRI of 0.1360 and CDTS of 0.3333 quantify the overall cross-domain variance and mean alignment, respectively.

The following directions are identified as high-priority for future investigation:

1) **Expanded RAG knowledge base**: The current RAG module relies on keyword matching over a static knowledge base of seven entries, leaving many prompts (e.g., orchestral, jazz, ocean waves, fire) to receive only generic conditioning. Expanding the corpus to cover 50+ acoustic concepts and replacing keyword matching with a dense neural retriever trained on audio-language correspondence data could substantially improve descriptor precision and recall.
2) **Multi-model CLAP ensemble**: The GIAF metrics are currently grounded in a single CLAP checkpoint. Exploring ensemble alignment scoring across multiple audio-language models (e.g., CLAP + LAION-CLAP + MERT) would improve robustness to biases in any single pre-trained encoder.
3) **Human perceptual validation**: Integration of human listener studies with GIAF scores would validate the degree to which automated intent-alignment metrics predict listener satisfaction in ecologically valid listening conditions.
4) **Statistical reliability**: Running multiple seeds per prompt (3–5 generations each) to compute confidence intervals for POAS scores would strengthen the statistical reliability of cross-domain findings.
5) **Larger-scale domain expansion**: Extension of the unified framework to larger sample sizes, more prompt diversity, and additional acoustic domains (e.g., bioacoustics, industrial, cinematic foley) would broaden the generalisability of GIAF-based evaluation.

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